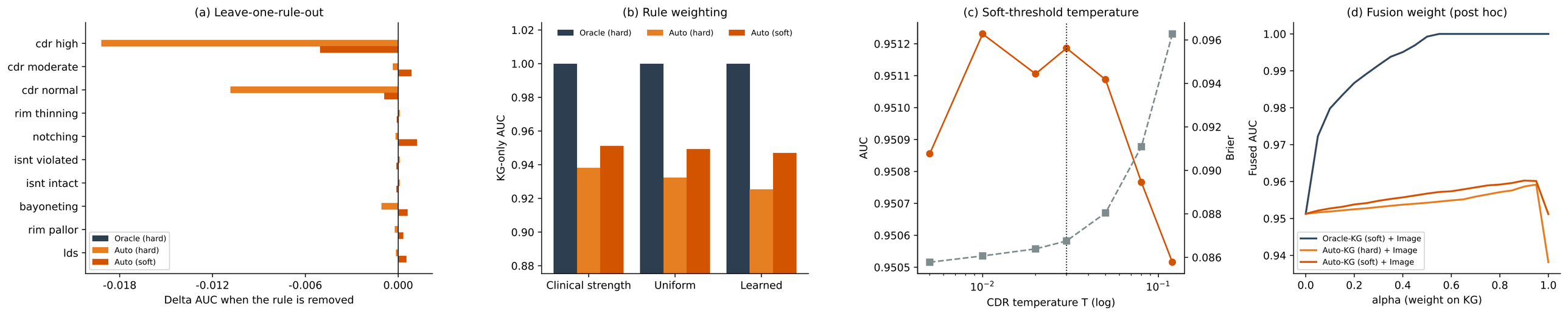
(a) Leave-one-rule-out
cdr high
cdr moderate
cdr normal
rim thinning
notching
isnt violated
isnt intact
bayoneting
rim pallor
lds
Oracle (hard)
Auto (hard)
Auto (soft)
-0.018
-0.012
-0.006
0.000
Delta AUC when the rule is removed
(b) Rule weighting
Oracle (hard)
Auto (hard)
Auto (soft)
KG-only AUC
1.02
1.00
0.98
0.96
0.94
0.92
0.90
0.88
Clinical strength
Uniform
Learned
(c) Soft-threshold temperature
AUC
0.9512
0.9511
0.9510
0.9509
0.9508
0.9507
0.9506
0.9505
Brier
0.096
0.094
0.092
0.090
0.088
0.086
10^{-2}
10^{-1}
CDR temperature T (log)
(d) Fusion weight (post hoc)
Fused AUC
1.00
0.99
0.98
0.97
0.96
0.95
0.94
Oracle-KG (soft) + Image
Auto-KG (hard) + Image
Auto-KG (soft) + Image
0.0
0.2
0.4
0.6
0.8
1.0
alpha (weight on KG)